

4.3 Photosynthesis, food production and populations

This section begins with a consideration of the biochemical pathways of photosynthesis and its importance as a synthetic process upon which most food chains and food webs are based.

Learners gain an appreciation of the role of microorganisms in recycling materials within the environment and maintaining balance within ecosystems. Microorganisms are also used as natural biosensors by environmental biologists to detect and measure the presence of chemical compounds in the environment.

In this section learners consider ecosystems on different scales from the small (the rumen of many herbivores) to the very large (farms required for global food production). The need to conserve environmental resources in a sustainable fashion is considered, whilst appreciating the potential conflict arising from the needs of an increasing human population. Learners also consider the impacts of human activities on the natural environment and biodiversity.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.